Google Ads API — Design Document

Applicant: Plottlife (handys Co., Ltd.) **Website:** https://life.plott.co.kr **Internal tool:** PLOTTLIFE-MKT Agent
Prepared for: Google Ads API Basic Access Application **Date:** 2026-04

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1. Executive Summary

Plottlife is a Korea-based short-term rental platform (1–20 week stays). We are building an internal marketing dashboard for our content team to plan SEO-driven blog articles. To understand search demand across Korean and English-speaking audiences, we query monthly search volume, competition, and country-level interest using:

1. **Naver Search Ads API** (already integrated) — Korean market
2. **Google Ads API — Keyword Planner Service** (this application) — English-speaking and overseas audiences

The Google Ads API token will be used **exclusively for read-only keyword research**. No Google Ads campaigns are created, edited, bid, or served through this token.

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2. Business Context

Our primary audiences include:

- International students (D-2 / D-4 visa holders)
- Corporate expats on short assignments
- Digital nomads and travelers on one-month stays
- Domestic residents in transitional housing situations

Many of these audiences search Google in English for terms such as `short term rental seoul`, `no deposit rental korea`, `student housing korea`. The Naver Search Ads API cannot surface these queries because Naver is Korean-language dominant and returns `< 10` impressions for English keywords.

The Google Ads Keyword Planner is the only data source that provides monthly search volume, competition level, and geographic breakdown for English queries targeting our market.

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3. High-Level Architecture

```
┌──────────────────────────────────────────────────────────┐
│            Internal Marketing Dashboard (Next.js)          │
│              (deployed at Vercel, HTTPS only)              │
└──────────────┬──────────────────────────────────┬──────────┘
               │                                  │
               │                                  │
               │ Server-side                      │ Server-side
               │ (Next.js Route Handlers)         │
               ▼                                  ▼
┌──────────────────────────┐          ┌──────────────────────────┐
│    Naver Search Ads      │          │   Google Ads API         │
│   (/keywordstool)        │          │  KeywordPlanIdeaService  │
│   Korean keywords        │          │  English keywords        │
└──────────────────────────┘          └──────────────────────────┘
               │                                  │
```

```
                └──────────────┬──────────────┘
                               │
                               ▼
                ┌──────────────────────────┐
                │    Supabase Postgres     │
                │  research_sources table  │
                │ (server access only, RLS)│
                └──────────────────────────┘
                               │
                               ▼
                ┌──────────────────────────┐
                │    Internal Dashboard    │
                │ /blog/research (Next.js) │
                │  Employee-only access    │
                └──────────────────────────┘
```

All Google Ads API calls occur **server-side** from our Next.js Route Handlers. API credentials are never exposed to browsers. Client-side code communicates only with our own `/api/*` endpoints and never with Google's endpoints directly.

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4. API Usage — Scope and Endpoints

4.1 Service and Method

We use a single Google Ads API service and method:

- **Service:** `KeywordPlanIdeaService`
- **Method:** `GenerateKeywordHistoricalMetrics` (and optionally `GenerateKeywordIdeas` for related-term expansion)

No other Google Ads services or methods are invoked. We **do not** use:

- `CampaignService` / `AdGroupService` / `AdService` (no campaign mgmt)
- `GoogleAdsService` mutate operations
- `ConversionActionService` / `RemarketingActionService`
- Customer creation endpoints

4.2 Request pattern

- Input: up to 20 Korean or English keywords per request
- Target language: `ko` (Korean) or `en` (English)
- Target geo constants: South Korea, plus supplementary markets (Vietnam, Indonesia, China, United States, Japan) for English queries
- Network: Google Search only
- `includeAdultKeywords` : false

4.3 Expected call volume

- **1,000 – 5,000 requests per month**
- Triggered manually by content team (roughly 20–50 keywords refreshed per week) and via a nightly cron job that refreshes top 100 seed keywords once every 7 days.
- Retry logic: exponential backoff (1s / 2s / 4s) on quota errors.

4.4 What we never do

- No campaign creation, editing, or serving

- No bid modification
- No data resale, redistribution, or bulk export to external parties
- No end-user-facing queries (all queries originate from internal staff)

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5. Data Handling & Storage

5.1 What we store

Field	Source
`keyword`	Request input
`monthly_search_volume`	Google Ads API response
`competition_level`	Google Ads API response
`top_of_page_bid_low` / `high` (optional)	Google Ads API response
`country_breakdown` (optional, from our own aggregation)	derived
`enriched_at` timestamp	server clock

5.2 Where we store it

- **Database:** Supabase (PostgreSQL), hosted on AWS `ap-northeast-2`.
- **Table:** `research_sources` with `kind = 'keyword'`.
- **Access:** Row Level Security enabled; direct access is blocked for the anonymous client. All reads/writes occur via our authenticated server API routes using the `service_role` key held only on the server.

5.3 Retention & deletion

- Keyword rows are refreshed in place (upsert), not appended.
- Historical metric snapshots are not retained unless explicitly requested by a team member.
- On account termination or Google's request, the team can purge all `research_sources` rows with a single SQL statement.

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6. Security & Access Control

6.1 Credentials handling

- Developer token, OAuth client secret, and refresh token are stored as **server-side environment variables** on Vercel (Production + Preview), never in git, never in browser bundles.
- Environment variables are **not** prefixed with `NEXT_PUBLIC_`, so Next.js excludes them from the client bundle at compile time.
- `src/lib/ads/google-ads.ts` wrapper imports `server-only`, which causes a build-time error if accidentally imported into a client component.

6.2 OAuth scope and token lifecycle

- OAuth scope: `https://www.googleapis.com/auth/adwords`
- Refresh token is obtained once via offline OAuth flow and stored server-side.
- No user-facing OAuth prompts (the tool has no third-party users).

6.3 User access to the dashboard

- The dashboard at `/blog/research` is available only on our internal Vercel deployment.
- All authentication to the dashboard is gated behind company SSO / internal credentials.
- No public API exposes Google Ads data.

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7. Compliance

We commit to comply with:

- **Google Ads API Policies and Required Minimum Functionality** (https://developers.google.com/google-ads/api/docs/policies/required-minimum-functionality) — As a read-only tool, the MRF requirements that apply to campaign management tools do not apply. However, we observe all general API policies (no resale, appropriate usage, attribution).
- **Google API Services User Data Policy** — We do not store any Google user data beyond the keyword metrics returned by the API.
- **Personal Information Protection Act (PIPA, Korea)** — No personal data is transmitted to or stored from the Google Ads API.
- **Terms and Conditions** accepted via the application form.

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8. Appendix — Sample request / response

Sample `KeywordPlanIdeaService.GenerateKeywordHistoricalMetrics` request (Node, `google-ads-api` client):

```
const customer = client.Customer({
  customer_id: process.env.GOOGLE_ADS_CUSTOMER_ID,
  login_customer_id: process.env.GOOGLE_ADS_LOGIN_CUSTOMER_ID,
  refresh_token: process.env.GOOGLE_ADS_REFRESH_TOKEN,
})

const result = await customer.keywordPlanIdeas.generateKeywordHistoricalMetrics({
  keywords: ["short term rental seoul", "no deposit rental korea"],
  language: "languageConstants/1000", // English
  geo_target_constants: ["geoTargetConstants/2410"], // South Korea
  include_adult_keywords: false,
})
// → [{ text: "short term rental seoul", keyword_metrics: { avg_monthly_searches,
competition, ... } }]
```

Contact

- API contact: julian.kim@handys.co.kr
- Company: handys Co., Ltd. (Plottlife)
- Website: https://life.plott.co.kr